Persistent activity pattern in large-scale recurrent network model of working memory validated *in vitro* with network clamp method

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Introduction

We have developed a method to record voltage responses in biological neurons *in vitro* with network model derived synaptic inputs. This interface involves injecting input received by a model neuron within a spiking network into a *in vitro* cell through intracellular patch clamp recording. This network clamp procedure allows the testing of model-driven hypotheses at the level of single-neuron dynamics. As a first example application, we introduce a novel spiking model of persistent activity and interface the network to pyramidal neurons from mouse PFC.

The Method

Network clamp allows for "*in vivo* like" stimulation *in vitro*

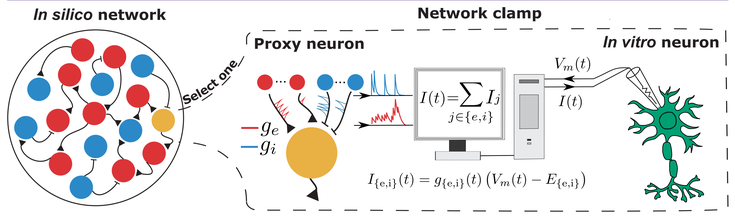


Fig 1. Network Clamp aims to bridge the gap between *in silico* and *in vitro* techniques by combining the unique perspective of network models with *in vitro* neurons in whole-cell patch clamp, which are difficult to access and control for *in vivo*. With our method, synaptic inputs derived from a neuron model are used to stimulate a neuron *in vitro*, thereby simulating the presence of an intact network.

Spiking neural network models propose a persistent activity mechanism

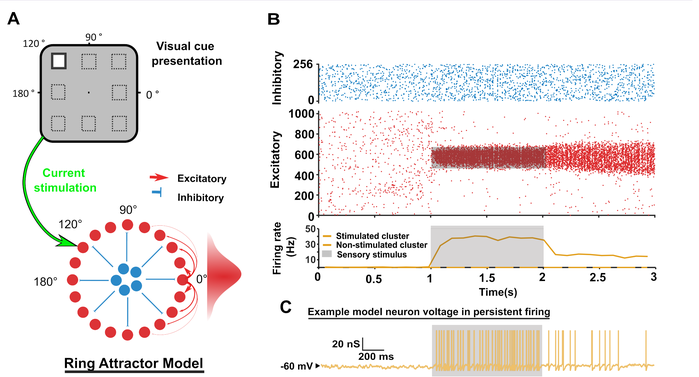


Fig 2. Ring attractor model exhibits persistent activity at realistic rates. **A.**The most established model of persistent activity is an attractor model representing a spatial working memory task where a fraction of excitatory neurons fire persistently to encode the position of a visual cue after the stimulus disappears. **B.**Activity of excitatory neurons (red) and inhibitory neurons (blue) arranged on the y-axis by their index number. The activity pattern is maintained after transient current stimulation representing sensory stimulus presentation. **C.**Example voltage trace from a persistent firing neuron model.

Network clamp experiment reveals unphysiological synaptic regime of established model

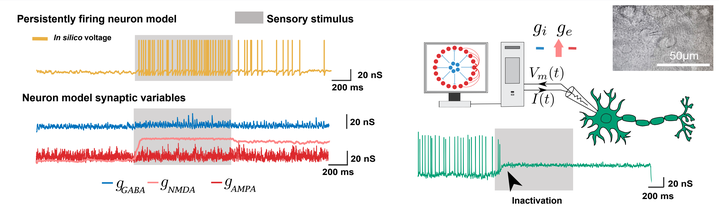


Fig 3. The ring attractor model shows an isolated elevation in NMDA conductance during persistent firing phase(mean = 28.9 nS, z-score = 25.62), while AMPA (mean=9nS,z-score=0.126) and GABA (mean=8 nS, z-score=0.767) remain relatively unchanged, both in both mean value and variability. This isolated NMDA response is not only biologically implausible but also creates an excitation-inhibition imbalance that is toxic for *in vitro* neurons, as the NMDA increase is not accompanied by a corresponding increase in inhibition. In the theoretical model, this imbalance is sufficient to prevent runaway excitation at the network level where a neuron's voltage reset after a spike; however, it causes inactivation in *in vitro* neurons.

Our Model

A spiking network model of persistent activity based on long-range clustered connectivity

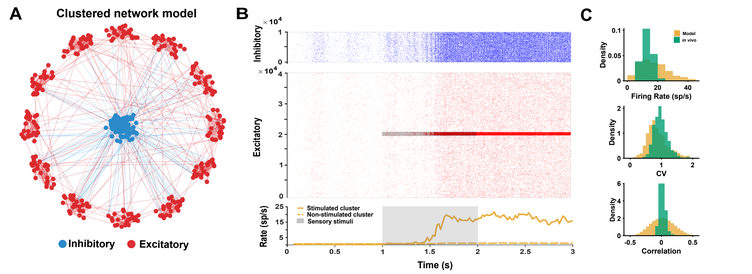


Fig 4. A.Network architecture. The spiking network is a large-scale model of PFC circuit. It is composed of $N = 50,000$ neurons, with $0.8N$ excitatory neurons and $0.2N$ inhibitory neurons. The key feature of the network topology is clustering amongst excitatory neurons divided into 40 clusters. Connections within a cluster occur with higher probability p_{in}, than connections across clusters p_{out}. **B. Spike Raster** . A transient current stimulus is given to a single cluster of excitatory neurons (grey square), which presents a bump of activity (17.4Hz) that persist after stimulus ends. **C. The model bump activity replicates key features observed in biological network activity recorded in the marmoset dlPFC during ODR task.**

Evoking model prediction in *in vitro* neuron

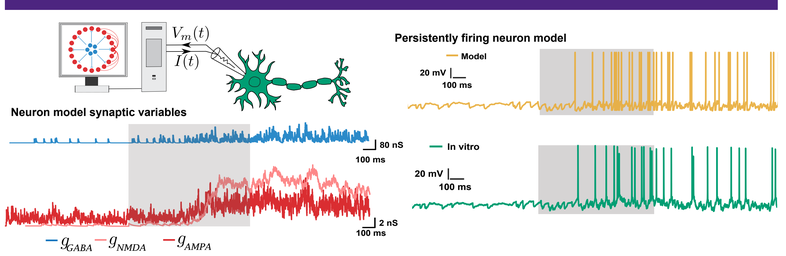


Fig 5. Our model evokes persistent activity *in vitro* close to prediction. Our model can evoke persistent activity into neuron *in vitro* because synaptic conductance present a more balanced cohesive dynamics that transiently hyperpolarize the neuron.

Our model evoke predicted response in different regime persistent firing, background activity, distractor

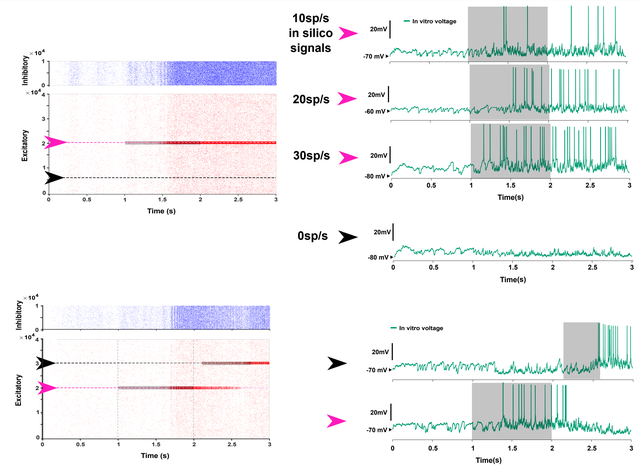


Fig 6. Injecting model-driven inputs via network clamp. **A.** Network clamp *in vitro* neuron response to inputs from model neurons with different firing rates within the persitent bump.(10,20,30,40 sp/s) and outside of the persistent bump (0sp/s). **B.** Network clamp evokes distractor driven dissipation of persistent activity.)

Network clamp can be used to study subthreshold dynamics effect of neuromodulators and other drugs

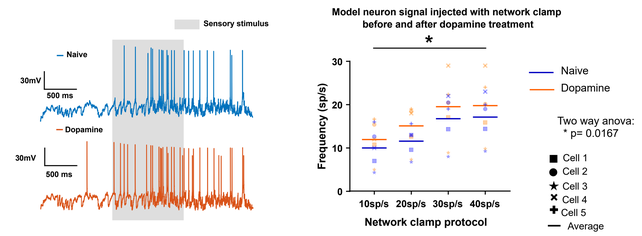


Fig 7. Dopamine increase excitability of mPFC mouse neurons in "*in vivo* like" context of network clamp. Dopamine is a neuromodulator crucial for working memory and persistent activity. Its action on single cell excitability is not yet well understood in the context of network to single cell interaction.

Conclusions

1. We developed a large-scale spiking network model of persistent activity based on clustered connectivity.
2. Injecting neuron model balanced conductance inputs to neuron *in vitro* evokes predicted response in wide array of contexts.

Acknowledgements

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